



MACHINE LEARNING ANALYSIS OF POVERTY DYNAMICS AND EURO ADOPTION IN EASTERN EUROPE

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THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
MASTER OF SCIENCE IN DATA SCIENCE & SOCIETY
AT THE SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
OF TILBURG UNIVERSITY

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DATE

May 2026

WORD COUNT

8799

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Abstract

This thesis examines the extent to which machine learning models can predict the at-risk-of-poverty rate using socioeconomic indicators within Eastern European countries in the context of Eurozone integration. Existing research on poverty vulnerability within Eastern Europe has focused primarily on descriptive macroeconomic analysis, while comparatively limited attention has been given to machine learning and explainable artificial intelligence techniques.

The study implemented and compared Linear Regression, Support Vector Regression, and Random Forest Regression models using a multidimensional socioeconomic dataset compiled primarily from Eurostat indicators covering Bulgaria, Croatia, Lithuania, and Poland between 2014 and 2025. The dataset included variables related to inequality, unemployment, inflation, GDP growth, housing burden, and household financial vulnerability. Bayesian hyperparameter optimisation and SHAP explainability analysis were additionally applied to improve predictive performance and interpretability.

The findings demonstrated that the optimised Support Vector Regression model achieved the strongest predictive performance, suggesting that poverty-related socioeconomic dynamics are characterised by important non-linear structural relationships. The explainability analysis further identified inequality, financial vulnerability, unemployment, and inflation-related indicators as influential predictors associated with poverty-related outcomes.

Overall, the study demonstrates that combining machine learning forecasting with explainable artificial intelligence techniques can provide valuable insight into multidimensional socioeconomic vulnerability within Eastern European economies.

Keywords: Machine Learning, Poverty, Euro Adoption, Eastern Europe, Random Forest, SHAP, Socioeconomic Inequality

o.1 *Data Source, Ethics, Code, and Technology Statement*

The datasets used in this thesis were obtained from publicly accessible sources, including Eurostat databases. The thesis did not involve collecting new data from human participants or animals. The data used are anonymised and publicly available for academic research purposes.

All figures, tables, and visualisations included in this thesis were created by the author unless otherwise stated. No copyrighted figures were reproduced without permission.

The code used for data preprocessing, exploratory data analysis, machine learning modelling, model evaluation, hyperparameter optimisation, and SHAP analysis was written by the author in Python. The primary libraries and frameworks used include pandas, numpy, matplotlib, seaborn, scikit-learn, shap, scipy, and statsmodels.

Zotero was used for reference management. Overleaf and LaTeX were used for thesis formatting and typesetting. Generative AI tools were used exclusively for language assistance, structural guidance, and academic writing support. All final analysis, interpretation, results, and conclusions were reviewed, edited, and approved by the author. The source code and implementation files used in this thesis are available at:

<https://github.com/Yoannie1311/Thesis>

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1 INTRODUCTION

1.1 *Research Background*

Despite continued economic integration within the European Union, socioeconomic inequality and financial vulnerability remain significant challenges across Eastern European countries. Differences in income distribution, labour market conditions, inflation exposure, and household financial resilience continue to affect living standards throughout the region, particularly in countries preparing for or recently entering the Eurozone. Within the European Union, one of the most commonly used indicators for measuring such vulnerability is the at-risk-of-poverty rate (AROP), which measures the proportion of individuals living below 60% of the national median equivalised disposable income (Mysíková & Želinský, 2019).

The at-risk-of-poverty rate is closely connected to broader socioeconomic conditions such as unemployment, inflation, economic growth, income inequality, housing-related financial burden, and the ability of households to manage unexpected expenses (Drishti et al., 2023; Whelan & Maître, 2012). In Eastern European countries, these dynamics are particularly important due to ongoing economic convergence processes and the structural adjustments associated with European monetary integration.

The adoption of the euro represents one of the most significant forms of economic integration within the European Union. Eurozone membership is generally associated with greater monetary stability, stronger macroeconomic coordination, and reduced exchange-rate uncertainty (Panaite & Radoi, 2025). However, concerns regarding inflation, purchasing power, and socioeconomic vulnerability remain common in countries undergoing or considering euro adoption (Soper, 2025). Previous comparative studies indicate that macroeconomic indicators such as GDP growth, inflation, unemployment, and inequality evolve differently across Eurozone and non-Eurozone countries in Eastern Europe (Borowski et al., 2025; University of Warmia and Mazury in Olsztyn et al., 2019).

At the same time, the increasing availability of socioeconomic data has contributed to growing interest in the use of machine learning techniques within economics and public policy research. Compared to traditional econometric approaches, machine learning models are better suited for identifying complex and non-linear relationships between multiple socioeconomic variables (Masini et al., 2023). Recent studies have demonstrated the potential of machine learning techniques for predicting poverty-related outcomes and financial vulnerability using large socioeconomic datasets (Zixi, 2021).

This thesis examines poverty-related socioeconomic dynamics in selected Eastern European countries between 2014 and 2025 using machine learning techniques. More specifically, the study analyses the relationship between the at-risk-of-poverty rate as a target variable and a set of socioeconomic indicators including unemployment, inflation, GDP growth, income inequality, housing burden, and financial vulnerability indicators. The predictive performance of Linear Regression, Support Vector Regression, and Random Forest Regression models is evaluated and interpreted using SHAP explainability methods in order to identify the variables most strongly associated with poverty-related dynamics in the context of Eurozone integration.

1.2 *Problem Statement and Research Goal*

Although economic integration within the European Union has contributed to economic growth and convergence across many Eastern European countries, substantial socioeconomic disparities remain visible throughout the region. Countries preparing for or recently entering the Eurozone continue to face challenges related to inflation sensitivity, inequality, labour market instability, and household financial vulnerability. At the same time, the socioeconomic consequences of euro adoption remain debated within both academic literature and public discourse (Borowski et al., 2025; Soper, 2025).

Existing research has extensively examined Eurozone integration from macroeconomic and political perspectives (Panait & Radoi, 2025; University of Warmia and Mazury in Olsztyn et al., 2019). However, comparatively limited attention has been given to the use of machine learning techniques for analysing poverty-related socioeconomic vulnerability within the context of Eurozone integration. In addition, many existing studies focus primarily on predictive performance without adequately addressing model interpretability, which remains particularly important in policy-oriented socioeconomic research (Lundberg & Lee, 2017).

The primary goal of this thesis is therefore to investigate whether machine learning methods can effectively analyse and predict the at-risk-of-poverty rate in Eastern European countries within the context of Eurozone integration.

1.3 *Research Objectives*

The main objective of this thesis is to analyse the relationship between Eurozone integration and poverty-related socio-economic indicators in Eastern European countries using machine learning techniques.

The specific objectives of the study are:

- To identify the socio-economic indicators most strongly associated with poverty and financial vulnerability in Eastern European countries.
- To compare the predictive performance of Linear Regression, Support Vector Regression, and Random Forest Regression models.
- To examine the relationship between euro adoption, inequality, inflation, unemployment, and material deprivation.
- To evaluate the usefulness of explainable artificial intelligence methods for interpreting socio-economic machine learning models.
- To contribute to existing literature by integrating machine learning prediction and explainable AI within the context of Eurozone integration research.

1.4 *Research Questions*

The main research question addressed in this thesis is:

To what extent can machine learning models predict the at-risk-of-poverty rate using socioeconomic indicators in Eastern European countries within the context of Eurozone integration?

The study additionally addresses the following sub-questions:

- RQ1: Which socioeconomic indicators are most strongly associated with the at-risk-of-poverty rate in Eastern European countries?
- RQ2: How do Linear Regression, Support Vector Regression, and Random Forest Regression compare in predictive performance?
- RQ3: How are inequality, inflation, and financial vulnerability associated with poverty-related outcomes in the context of Eurozone integration?
- RQ4: How can SHAP explainability methods improve the interpretability of machine learning models in socioeconomic research?
- RQ5: How robust are the machine learning models when evaluated on unseen time-series data?

2 LITERATURE REVIEW

2.1 *Poverty and Socio-Economic Inequality*

Poverty is increasingly recognised as a multidimensional phenomenon extending beyond insufficient income alone. Contemporary research emphasises that poverty additionally involves limited access to healthcare, education, employment opportunities, housing security, and adequate living standards (Maggino, 2023; Usmanova et al., 2022; Zixi, 2021). This broader understanding emerged partly in response to criticism of traditional income-based poverty measurements, which may fail to capture material deprivation and social exclusion experienced by households whose income exceeds official poverty thresholds (Mysíková & Želinský, 2019; Whelan & Maître, 2012). Consequently, poverty within the European context is increasingly analysed using multidimensional frameworks integrating economic, social, and material indicators.

The distinction between absolute and relative poverty remains central within poverty research. Absolute poverty refers to the inability to satisfy basic human needs, whereas relative poverty evaluates deprivation in relation to broader societal living standards and social participation (Maggino, 2023). Within the European Union, relative poverty measures such as the at-risk-of-poverty rate (AROP) are widely used because they better reflect socio-economic exclusion within developed economies (Mysíková & Želinský, 2019). The AROP indicator measures the proportion of individuals living below 60% of the national median equivalised disposable income and is frequently analysed alongside indicators of material deprivation and financial vulnerability (De Vos & Zaidi, 1997; Mysíková & Želinský, 2019).

Existing literature identifies a strong relationship between inequality and socio-economic vulnerability. Higher inequality levels are associated with greater material deprivation and weaker economic resilience among lower-income households (Van De Werfhorst & Salverda, 2012). Comparative analyses across European Union member states similarly demonstrate that countries characterised by higher Gini coefficient values often experience greater poverty exposure and lower average living standards (Sompolska-Rzechuła & Kurdyś-Kujawska, 2022). Although GDP per capita remains connected to quality of life and poverty reduction, economic growth alone does not necessarily eliminate socio-economic disparities (Sompolska-Rzechuła & Kurdyś-Kujawska, 2022).

Within Eastern Europe, poverty and inequality remain closely connected to post-transition restructuring and uneven convergence within the European Union. Although EU accession contributed to economic growth

and macroeconomic integration, disparities in income distribution, labour market conditions, and living standards continue to persist (Kotseva-Tikova & Dvorak, 2025). Bulgaria in particular continues to record some of the highest inequality levels and lowest income indicators within the European Union (Kotseva-Tikova & Dvorak, 2025). OECD analyses similarly indicate that several Eastern European economies remain vulnerable to unemployment, housing affordability pressures, and household financial insecurity despite broader convergence trends (OECD, 2024, 2025).

Financial vulnerability is increasingly analysed in relation to inflation sensitivity and household resilience. Lower-income households are generally more exposed to inflationary pressure because a larger share of expenditures is allocated toward essential goods and services (OECD, 2024). Research focusing on Eastern European economies suggests that inflation-related concerns are associated with declining financial security and reduced ability to absorb unexpected economic shocks (Kotseva-Tikova & Dvorak, 2025; Soper, 2025). Consequently, indicators such as unemployment, housing burden, inability to face unexpected expenses, and material deprivation are increasingly analysed together rather than relying exclusively on income-based poverty measures.

The literature additionally emphasises the importance of structural and institutional factors influencing poverty outcomes. Labour market conditions, institutional quality, welfare systems, and unequal access to opportunities significantly shape socio-economic vulnerability and social mobility (Maggino, 2023; Sompolska-Rzechuła & Kurdyś-Kujawska, 2022). Political and institutional perspectives similarly suggest that social protection systems and governmental policies play an important role in reducing poverty exposure and improving household resilience during periods of economic adjustment (OECD, 2025).

2.2 *Eurozone Integration, Convergence, and Socio-Economic Vulnerability*

Comparative empirical studies provide useful benchmarks for understanding how Eurozone integration has affected Central and Eastern European economies. Research comparing newer Eurozone members with non-Euro countries suggests that the effects of euro adoption differ depending on structural conditions, institutional preparedness, and broader macroeconomic resilience (Panait & Radoi, 2025; University of Warmia and Mazury in Olsztyn et al., 2019). Heller and Warzala observe that several Eurozone members experienced stronger macroeconomic stabilisation and convergence following euro adoption, although membership alone did not fully protect economies from external shocks during crisis periods (University of Warmia and Mazury in Olsztyn et al., 2019).

More recent analyses similarly indicate that Eurozone countries displayed lower interest-rate volatility and greater monetary stability during the inflationary period of 2022–2023 (Panait & Radoi, 2025). Panait and Radoi report that long-term interest rates in non-Euro countries such as Romania and Hungary exceeded 6–7%, whereas several Eurozone countries maintained substantially lower borrowing costs and volatility (Panait & Radoi, 2025). Their findings additionally demonstrate substantial GDP convergence among several Eastern European economies between 2005 and 2023 (Panait & Radoi, 2025). At the same time, the authors note that non-Euro countries remained more exposed to inflationary pressure, financing costs, and structural uncertainty.

These studies provide an important benchmark for the present research. While existing comparative analyses primarily focus on macroeconomic convergence and monetary stability, the present thesis extends this discussion by examining whether socioeconomic indicators can be used to predict the at-risk-of-poverty rate through machine learning methods.

2.3 *Machine Learning in Socio-Economic Research*

The increasing availability of large socio-economic datasets has contributed to growing interest in the application of machine learning methods within economics and public policy research. Existing studies suggest that machine learning approaches are particularly useful for analysing multidimensional socio-economic outcomes characterised by complex and potentially non-linear relationships between variables (Masini et al., 2023). As a result, machine learning techniques are increasingly applied in research examining poverty, inequality, unemployment, financial vulnerability, and broader socio-economic forecasting problems.

Recent literature demonstrates substantial growth in the use of machine learning methods for poverty prediction and socio-economic forecasting (Usmanova et al., 2022; Zixi, 2021). Poverty-related outcomes are shaped by interconnected economic, demographic, labour market, and institutional factors that may be difficult to capture using traditional linear approaches alone (Zixi, 2021). Machine learning models therefore offer greater flexibility for identifying complex relationships across multidimensional datasets and evolving socio-economic conditions (Masini et al., 2023).

Existing poverty prediction research increasingly incorporates multidimensional indicators rather than relying exclusively on income-based measures. Researchers frequently combine labour market conditions, demographic characteristics, housing-related indicators, education variables, healthcare access, and measures of financial vulnerability (Usmanova et al., 2022; Zixi, 2021). Studies additionally utilise survey-based data, admin-

istrative records, and alternative data sources to analyse socio-economic vulnerability and regional inequality patterns (Usmanova et al., 2022).

Within socio-economic forecasting research, studies increasingly compare multiple machine learning approaches in order to evaluate predictive performance across different economic environments and datasets. At the same time, interpretability remains an important challenge because models achieving stronger predictive performance may provide less transparent explanations regarding the influence of individual socio-economic variables (Lundberg & Lee, 2017). The literature additionally highlights the growing importance of optimisation techniques within machine learning research (Masini et al., 2023).

2.4 *Explainable Artificial Intelligence in Economics*

Despite the growing use of machine learning within economics and public policy research, concerns regarding model interpretability remain significant. Advanced predictive models frequently achieve strong predictive performance while providing limited transparency regarding the relationships driving their outputs. Within socio-economic research, this creates important challenges because policy-oriented analysis requires not only accurate predictions but also interpretable understanding of the socio-economic factors associated with those predictions (Lundberg & Lee, 2017; Masini et al., 2023).

The increasing complexity of predictive systems contributed to the emergence of Explainable Artificial Intelligence (XAI), a field focused on improving the transparency and interpretability of machine learning models. Existing literature emphasises that explainability techniques are particularly important within applied economic and socio-economic research environments where model outputs may influence policy evaluation and social decision-making processes (Lundberg & Lee, 2017).

Among the most widely used explainability approaches is SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017), particularly relevant within poverty and inequality research where socio-economic indicators are highly interconnected and difficult to interpret using traditional statistical techniques alone.

Studies comparing multiple machine learning approaches frequently report that models achieving stronger predictive performance may simultaneously become more difficult to interpret directly (Masini et al., 2023). Explainability techniques therefore help address the trade-off between predictive accuracy and transparency by providing systematic approaches for analysing model behaviour and variable influence.

Although explainable AI has developed rapidly in recent years, its application within socio-economic and Eurozone-related research remains comparatively limited. Existing studies frequently prioritise predictive accuracy or model comparison without systematically incorporating interpretability frameworks capable of identifying the socio-economic variables most strongly associated with predictive outcomes (Usmanova et al., 2022; Zixi, 2021).

2.5 *Research Gap*

Existing research provides extensive discussion on poverty, socio-economic inequality, European economic integration, and machine learning methodologies (Sompolska-Rzechuła & Kurdyś-Kujawska, 2022; Van De Werfhorst & Salverda, 2012; Whelan & Maître, 2012). Similarly, machine learning research has increasingly demonstrated the advantages of advanced predictive methods for analysing complex socio-economic datasets and forecasting poverty-related outcomes (Usmanova et al., 2022; Zixi, 2021). However, the literature remains fragmented across these research domains.

Research examining Eurozone integration continues to rely predominantly on traditional macroeconomic or econometric approaches focused on inflation, fiscal convergence, and monetary stability (Kotseva-Tikova & Dvorak, 2025; Soper, 2025). Comparatively limited attention has been given to the application of machine learning techniques for analysing poverty-related socio-economic dynamics associated with Eurozone integration, particularly within Eastern European countries.

At the same time, many machine learning poverty studies prioritise predictive performance without integrating broader economic integration perspectives (Usmanova et al., 2022; Zixi, 2021). Existing studies also rarely combine predictive modelling with explainability frameworks capable of identifying the socio-economic variables most strongly associated with poverty-related outcomes (Lundberg & Lee, 2017). This creates limitations for policy-oriented research where understanding the influence of individual variables is essential for meaningful interpretation and decision-making.

Another limitation concerns the relatively limited focus on Eastern European economies undergoing processes of economic convergence and Eurozone integration. Existing research frequently analyses broader EU-level trends or Western European economies, leaving a comparative gap regarding poverty vulnerability and socio-economic resilience within countries such as Bulgaria, Croatia, Lithuania, and Poland (Kotseva-Tikova & Dvorak, 2025).

This study attempts to address these limitations by combining socio-economic analysis of Eurozone integration with machine learning prediction methods and explainable artificial intelligence techniques. By applying Linear Regression, Support Vector Regression, and Random Forest models alongside SHAP explainability analysis, the study aims to provide both predictive insight and interpretable understanding of poverty-related socio-economic dynamics within Eastern European economies.

3 METHODOLOGY AND EXPERIMENTAL SETUP

3.1 *Research Methodology Overview*

The methodological framework of this study combines socio-economic analysis, predictive modelling, and explainable artificial intelligence techniques in order to examine relationships between the at-risk-of-poverty rate and a set of macroeconomic and socio-economic indicators.

The analysis focuses on Bulgaria, Croatia, Lithuania, and Poland between 2014 and 2025. The selected variables capture multiple dimensions of socio-economic vulnerability, including unemployment, inflation, income inequality, housing burden, financial insecurity, and Eurozone membership status.

The methodological process included dataset construction, preprocessing, exploratory data analysis, machine learning implementation, hyperparameter optimisation, model evaluation, and explainability analysis using SHAP. Linear Regression, Support Vector Regression, and Random Forest Regression models were implemented and compared in order to evaluate predictive performance across different modelling approaches.

To preserve the temporal structure of the data and avoid information leakage, the dataset was divided chronologically rather than randomly. Observations from 2014–2022 were used for model training, while observations from 2023–2025 were reserved for testing and out-of-sample evaluation.

A methodological flowchart illustrating the complete research pipeline is presented in Figure 1.

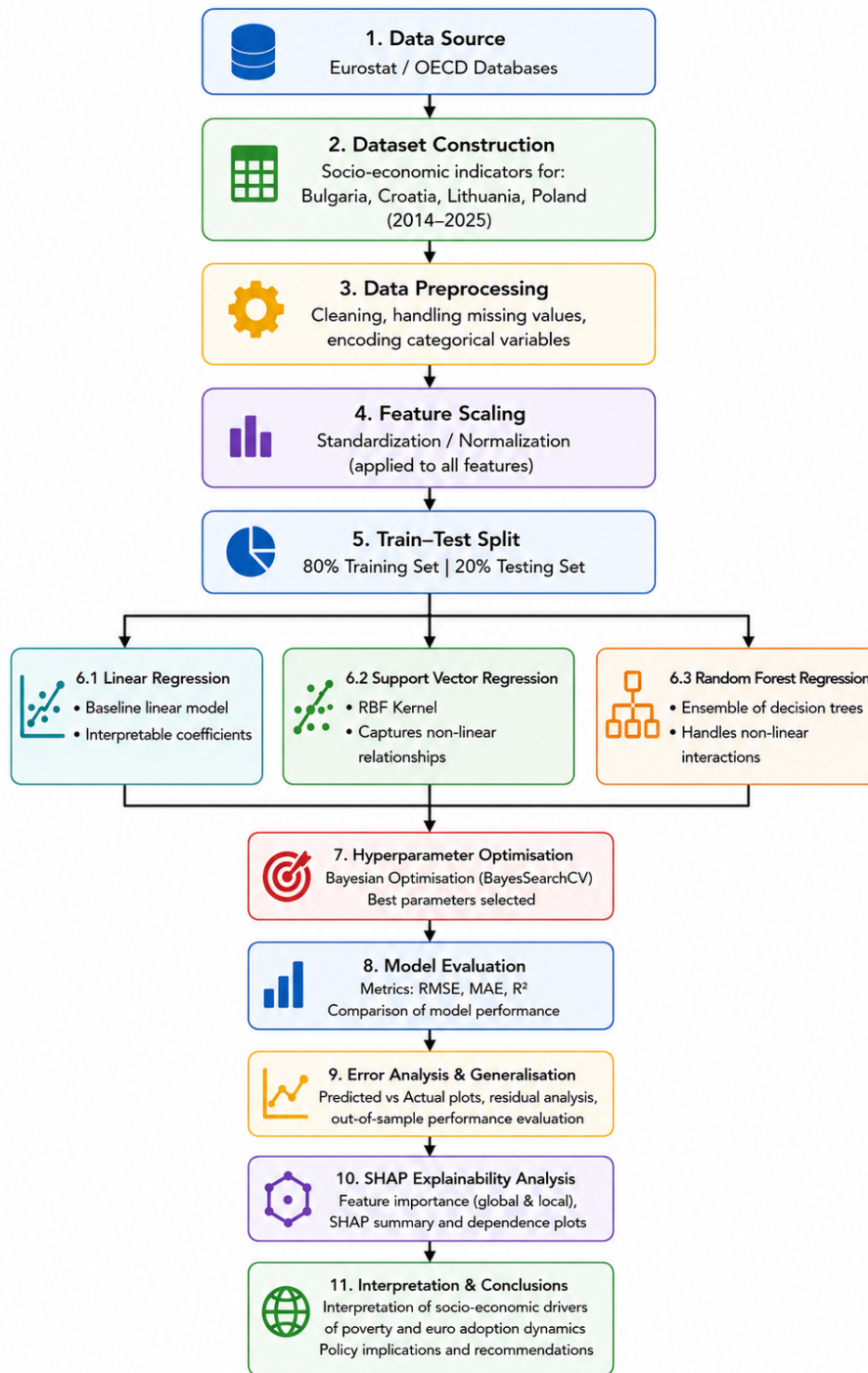


Figure 1: Overview of the research methodology and machine learning pipeline used in this study.

3.2 *Data Collection and Dataset Construction*

The dataset used in this study was manually constructed using publicly available socio-economic indicators obtained from Eurostat databases. The final dataset combined indicators associated with poverty, inequality, inflation, labour market conditions, housing burden, and financial vulnerability for Bulgaria, Croatia, Lithuania, and Poland between 2014 and 2025.

The selected countries represent different stages of Eurozone integration and economic convergence within Eastern Europe. Lithuania adopted the euro in 2015, Croatia joined the Eurozone in 2023, while Bulgaria and Poland continued using national currencies during the observed period. This allowed comparison across both Eurozone and non-Eurozone economies within a shared regional context.

The final dataset consisted of 48 country-year observations and included variables related to poverty, unemployment, inflation, GDP growth, income inequality, housing burden, financial insecurity, and euro adoption status. The at-risk-of-poverty rate was selected as the target variable because it represents one of the most widely used indicators of socio-economic vulnerability within European social policy research and allows consistent comparison across countries and time periods.

Because the indicators originated from multiple datasets, the data integration process required harmonisation of variable names, country identifiers, temporal coverage, and formatting structures before merging the datasets into a unified analytical framework. All variables were aligned on a yearly country-level basis prior to preprocessing and model implementation.

The analysis was implemented in Python using libraries including pandas, NumPy, scikit-learn, SHAP, matplotlib, seaborn, and scikit-optimize.

3.3 *Variable Selection and Definition*

The variables included in this study were selected based on previous literature in order to capture multiple dimensions of socio-economic conditions associated with poverty exposure, including labour market instability, inflationary pressure, financial insecurity, housing-related vulnerability, and macroeconomic performance.

Table 1 presents the variables included in the analytical dataset, their definitions, and sources.

Table 1: Variables included in the analytical dataset

Variable	Definition	Source
Poverty	At-risk-of-poverty rate after social transfers (%)	Eurostat
Unemployment	Annual unemployment rate (% of labour force)	Eurostat
Inflation	Annual inflation rate based on consumer prices (%)	Eurostat/OECD
GDP Growth	Annual GDP growth rate (%)	Eurostat/OECD
Gini	Gini coefficient of equivalised disposable income	Eurostat
Housing Burden	Housing cost overburden rate (%)	Eurostat
Unexpected Expenses	Inability to face unexpected financial expenses (%)	Eurostat
EuroAdopted	Binary indicator of Eurozone membership status	Constructed variable

The explanatory variables were selected because previous research frequently associates them with poverty vulnerability and socio-economic resilience within Eastern European economies. Unemployment and GDP growth capture labour market conditions and macroeconomic performance, while inflation reflects purchasing power and cost-of-living pressure. The Gini coefficient represents income inequality, whereas housing burden and inability to face unexpected expenses capture household-level financial vulnerability and economic insecurity.

The EuroAdopted variable was manually constructed as a binary indicator distinguishing Eurozone and non-Eurozone observations. Lithuania was coded as a Eurozone member from 2015 onward, while Croatia was coded as a Eurozone member from 2023 onward. Bulgaria and Poland remained coded as non-Eurozone economies throughout the observed period.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine relationships between variables, identify socio-economic trends, and evaluate differences between Eurozone and non-Eurozone countries within the dataset. Descriptive statistics and visual analysis were used to assess temporal patterns, variable interactions, and associations with the target variable.

Figure 2 presents the evolution of the at-risk-of-poverty rate across Bulgaria, Croatia, Lithuania, and Poland between 2014 and 2025, while additionally highlighting the timing of euro adoption in Lithuania (2015) and Croatia (2023).

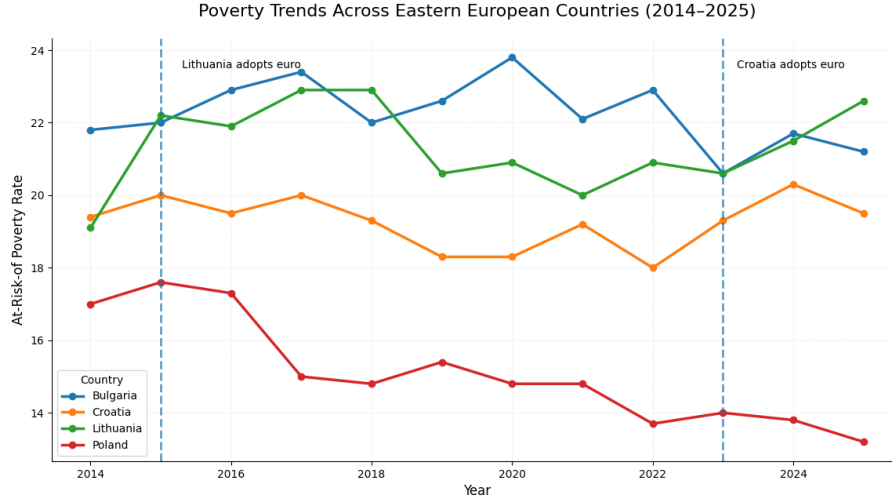


Figure 2: Poverty trends across Eastern European countries between 2014 and 2025.

The descriptive trends indicate substantial variation in poverty levels across countries. Bulgaria consistently recorded the highest poverty rates, while Poland displayed comparatively lower poverty exposure and a stronger downward trend over time. Lithuania and Croatia demonstrated more moderate poverty dynamics, although fluctuations remained visible across several years. The visualisations additionally suggest that Euro-zone membership alone was not associated with immediate or uniform reductions in poverty-related vulnerability.

Figure 3 illustrates the evolution of the selected socio-economic variables by country following variable normalisation and additionally indicates the chronological train-test split used for machine learning implementation.

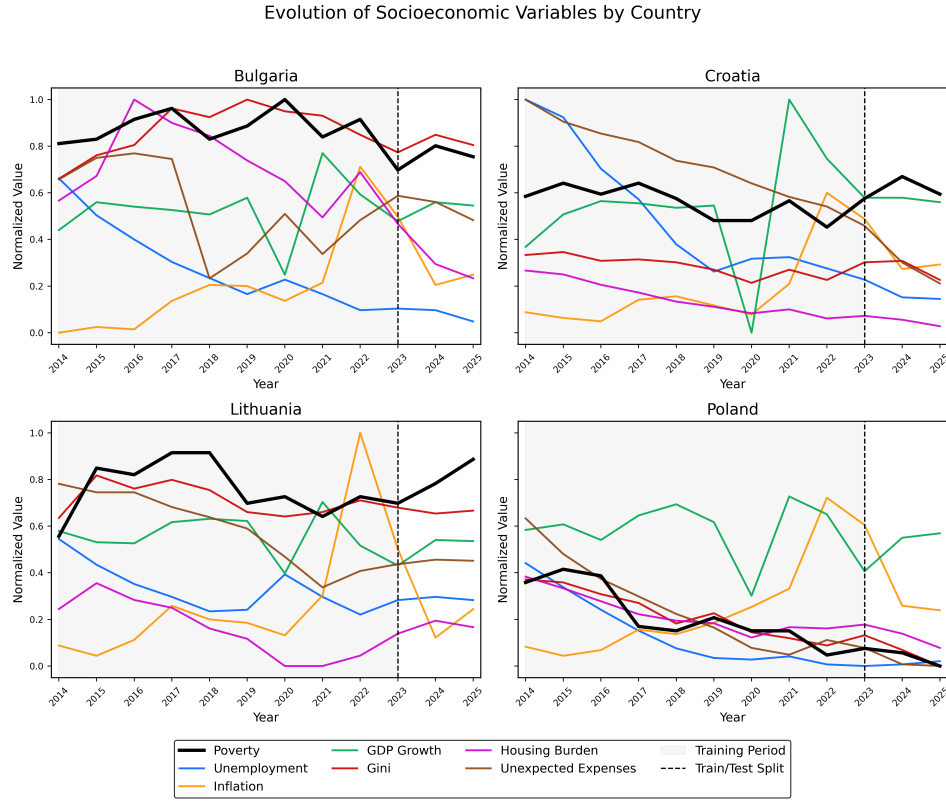


Figure 3: Evolution of socio-economic variables by country between 2014 and 2025.

The visualisations demonstrate substantial differences in socio-economic trajectories across countries and variables. Unemployment generally declined throughout the observed period, while inflation exhibited sharper fluctuations, particularly after 2020. Housing burden and inability to face unexpected expenses displayed persistent variation across countries, reflecting differences in household-level financial vulnerability and economic resilience.

Table 2 presents descriptive statistics for the socio-economic variables included in the analytical dataset.

Table 2: Descriptive statistics of socio-economic variables

Variable	Mean	Std	Min	Max
Poverty	19.49	2.98	13.20	23.80
Unemployment	6.89	3.23	2.80	17.30
Inflation	3.28	4.29	-1.60	18.90
GDP_Growth	3.20	3.03	-8.30	12.60
Gini	32.98	4.67	24.90	40.80
Housing_Burden	7.76	4.44	2.70	20.70
Unexpected_Expenses	42.63	10.44	22.50	63.70

The descriptive statistics indicate substantial variation across several socio-economic indicators, particularly inflation, unemployment, and financial vulnerability measures. This variability supports the use of comparative predictive modelling approaches capable of capturing heterogeneous socio-economic dynamics across countries and time periods.

To further examine relationships between variables, a correlation matrix was constructed using Pearson correlation coefficients, as presented in Figure 4.

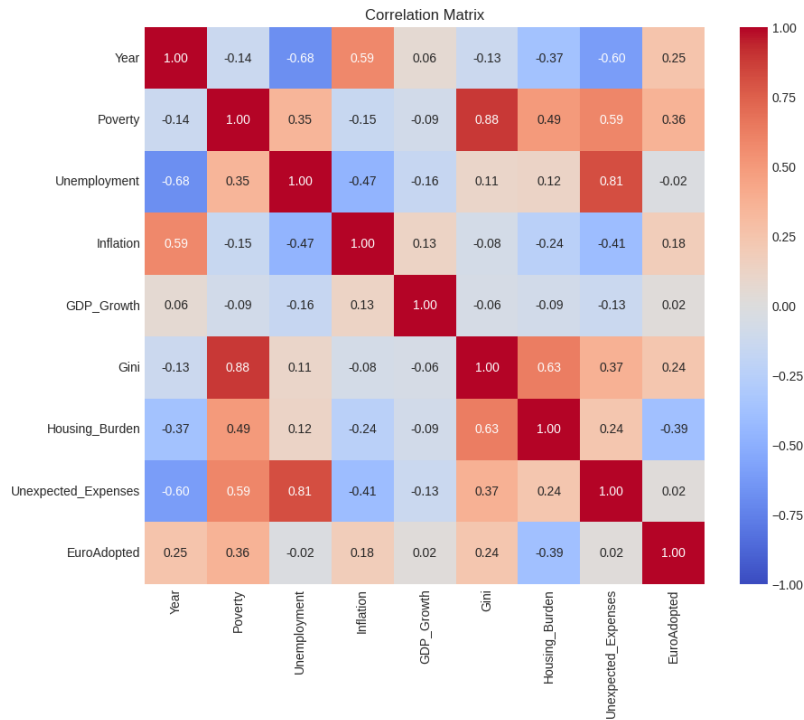


Figure 4: Correlation matrix of socio-economic variables included in the dataset.

The correlation analysis indicates that poverty demonstrated the strongest positive association with the Gini coefficient (0.88), suggesting a substantial relationship between income inequality and poverty vulnerability within the observed countries. Positive correlations were also observed between poverty and inability to face unexpected expenses (0.59), as well as housing burden (0.49), reflecting the importance of household-level financial insecurity within broader poverty dynamics. GDP growth demonstrated a comparatively weak negative relationship with poverty (-0.09), while inflation displayed limited direct correlation with the target variable during the observed period.

Overall, the exploratory analysis supports the multidimensional nature of poverty-related vulnerability and indicates that poverty outcomes are associated with multiple interconnected socio-economic factors rather than a single macroeconomic indicator alone.

3.5 *Machine Learning Methodology*

Three regression-based machine learning approaches were implemented in order to compare predictive performance across models characterised by different levels of complexity and interpretability: Linear Regression, Support Vector Regression (SVR), and Random Forest Regression.

The analysis was conducted in multiple stages. First, baseline versions of all models were implemented in order to establish initial predictive performance and provide a reference point for later optimisation. Hyperparameter optimisation was subsequently applied to Support Vector Regression and Random Forest Regression using Bayesian optimisation techniques in order to improve predictive performance and generalisation on unseen data. The final comparative analysis presented in this thesis is based on the optimised model configurations, while Linear Regression was retained as a non-optimised baseline benchmark.

Support Vector Regression and Random Forest Regression were selected because previous literature suggests that these approaches perform effectively when modelling complex and potentially non-linear socio-economic relationships.

Prior to model implementation, the dataset was inspected for missing values and formatting inconsistencies resulting from the integration of multiple socio-economic datasets. Variable names, country identifiers, and temporal structures were standardised during preprocessing to ensure consistency across observations.

Categorical country identifiers were retained for descriptive and visual analysis but excluded from numerical model training. The EuroAdopted

variable was encoded numerically in order to distinguish Eurozone and non-Eurozone observations within the machine learning framework.

To preserve the temporal structure of the dataset and avoid information leakage, the data was divided chronologically rather than randomly. Observations from 2014–2022 were used for model training, while observations from 2023–2025 were reserved for testing and out-of-sample evaluation. The training subset contained 36 observations, while the testing subset contained 12 observations.

Feature scaling was applied for Linear Regression and Support Vector Regression using standardisation techniques because Support Vector Regression is sensitive to differences in variable magnitude. Random Forest Regression did not require scaling because tree-based models are generally less sensitive to feature scale differences.

The machine learning analysis was implemented using Python libraries including scikit-learn and scikit-optimize.

3.6 *Hyperparameter Optimisation*

Optimisation was implemented through the BayesSearchCV framework from the scikit-optimize library. The optimisation process evaluated multiple parameter combinations for each model using cross-validation on the training dataset.

For Support Vector Regression, optimisation focused on the penalty parameter (C), kernel coefficient (γ), and epsilon margin. For Random Forest Regression, optimisation included parameters related to model complexity and tree structure, including the number of estimators, maximum tree depth, minimum samples split, minimum samples leaf, and maximum features.

The final optimised parameter configurations were selected based on the lowest Root Mean Squared Error (RMSE) values obtained during the optimisation process. These configurations were subsequently used for all final model evaluation, forecasting, and explainability analysis presented in the thesis.

The final hyperparameter configurations obtained through Bayesian optimisation are presented in the Results chapter.

3.7 *Model Evaluation, Explainability, and Statistical Analysis*

The predictive performance of the machine learning models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics provide complementary

perspectives on predictive accuracy, average prediction error, sensitivity to larger errors, and explained variance.

Model evaluation was conducted using the predefined chronological testing subset consisting of observations from 2023–2025 in order to evaluate model generalisation and reduce the risk of overfitting. Comparative evaluation was performed across Linear Regression, optimised Support Vector Regression, and optimised Random Forest Regression models.

In addition to numerical evaluation metrics, several visualisations were generated to examine model behaviour and forecasting performance. Comparative train-versus-test evaluation plots and country-level predicted-versus-actual visualisations were used to assess forecasting accuracy, error distribution, and generalisation across countries and time periods.

To improve model interpretability, SHAP (SHapley Additive exPlanations) analysis was applied to the optimised Random Forest Regression model using the SHAP TreeExplainer framework. SHAP values were used to estimate the contribution of individual variables to model predictions and identify the socio-economic indicators most strongly associated with poverty-related outcomes.

Alongside machine learning evaluation, the study included statistical comparison of socio-economic indicators before and after euro adoption. The statistical analysis focused primarily on Croatia because sufficient post-adoption observations were available following Eurozone accession in 2023. Independent sample t-tests were conducted to examine whether selected socio-economic indicators differed significantly before and after euro adoption.

Lithuania was not included in the t-test analysis because only one pre-adoption year was available within the observed dataset period, limiting the reliability of comparative statistical testing. The statistical analysis therefore served primarily as a complementary exploratory component supporting the broader machine learning evaluation of Eurozone integration and socio-economic vulnerability.

4 RESULTS AND ANALYSIS

4.1 *Baseline and Optimised Model Performance*

Three regression-based machine learning models were evaluated in this study: Linear Regression, Support Vector Regression (SVR), and Random Forest Regression. Following the implementation of the baseline models, Bayesian hyperparameter optimisation was applied to the Support Vector Regression and Random Forest models in order to improve predictive performance and model generalisation on unseen data.

Table 3 presents the hyperparameter ranges explored during Bayesian optimisation before selecting the final model configurations.

Table 3: Hyperparameter search space used during Bayesian optimisation

Model	Hyperparameter	Explored Range / Values
Optimised SVR	kernel	RBF
	C	0.1–100
	gamma	0.001–10
	epsilon	0.01–1.0
Optimised RF	n_estimators	50–300
	max_depth	2–20
	min_samples_split	2–10
	min_samples_leaf	1–5
	max_features	sqrt, log2

These ranges were evaluated using BayesSearchCV with time-series cross-validation. The optimisation procedure used 30 iterations for SVR and 20 iterations for Random Forest, with R^2 as the scoring metric.

Table 4 presents the final hyperparameter configurations selected through Bayesian optimisation for the Support Vector Regression and Random Forest models.

Table 4: Final optimised hyperparameter configurations

Model	Parameter	Value
Optimised SVR	C	8.4651
	epsilon	0.01
	gamma	0.0206
	kernel	RBF
Optimised RF	max_depth	8
	max_features	sqrt
	min_samples_leaf	1
	min_samples_split	2
	n_estimators	50

The optimised SVR configuration selected a relatively high C parameter together with a low epsilon value, indicating that the model prioritised minimising prediction error while remaining sensitive to smaller variations within the socio-economic data. The selected gamma parameter additionally allowed the Radial Basis Function kernel to capture more complex

non-linear relationships between the explanatory variables and poverty outcomes.

For the Random Forest model, the optimisation process selected a moderate tree depth together with a relatively limited number of estimators. This configuration reduced excessive model complexity while still allowing the model to capture non-linear interactions between socio-economic variables. The selected hyperparameters therefore contributed to balancing predictive performance and generalisation capability on unseen observations.

Overall, the Bayesian optimisation process improved both non-linear models by identifying parameter combinations that reduced prediction error while maintaining comparatively stable forecasting behaviour across the testing dataset.

Table 5 presents the comparative predictive performance of the final optimised models on the testing dataset.

Table 5: Comparative performance of final regression models on the testing dataset

Model	R ²	MAE	RMSE
Linear Regression	0.6593	1.4702	1.8776
Optimised Support Vector Regression	0.8089	1.0821	1.4063
Optimised Random Forest Regression	0.7757	1.2289	1.5233

The results demonstrate that the optimised Support Vector Regression model achieved the strongest overall predictive performance, obtaining the highest R² score together with the lowest MAE and RMSE values. This indicates that the optimised SVR model was most effective in capturing the complex relationships between the socio-economic indicators and the at-risk-of-poverty rate within the analysed Eastern European countries.

The optimised Random Forest model also produced strong predictive performance and substantially outperformed the baseline Random Forest implementation following Bayesian hyperparameter optimisation. The improvement in predictive accuracy suggests that optimisation contributed positively to model stability and generalisation performance on unseen observations.

Although Linear Regression produced comparatively weaker predictive performance, the model remained valuable for interpretability purposes because the coefficient structure allows direct examination of the influence exerted by individual socio-economic variables on poverty predictions.

Overall, the stronger performance of the optimised non-linear models suggests that poverty-related socio-economic dynamics within Eastern

European economies are characterised by partially non-linear interactions that cannot be fully captured through purely linear modelling approaches.

4.2 Forecasting Performance and Train-Test Generalisation

Figure 5 presents the forecasting performance of the implemented machine learning models across both the training and testing periods for the analysed Eastern European countries.

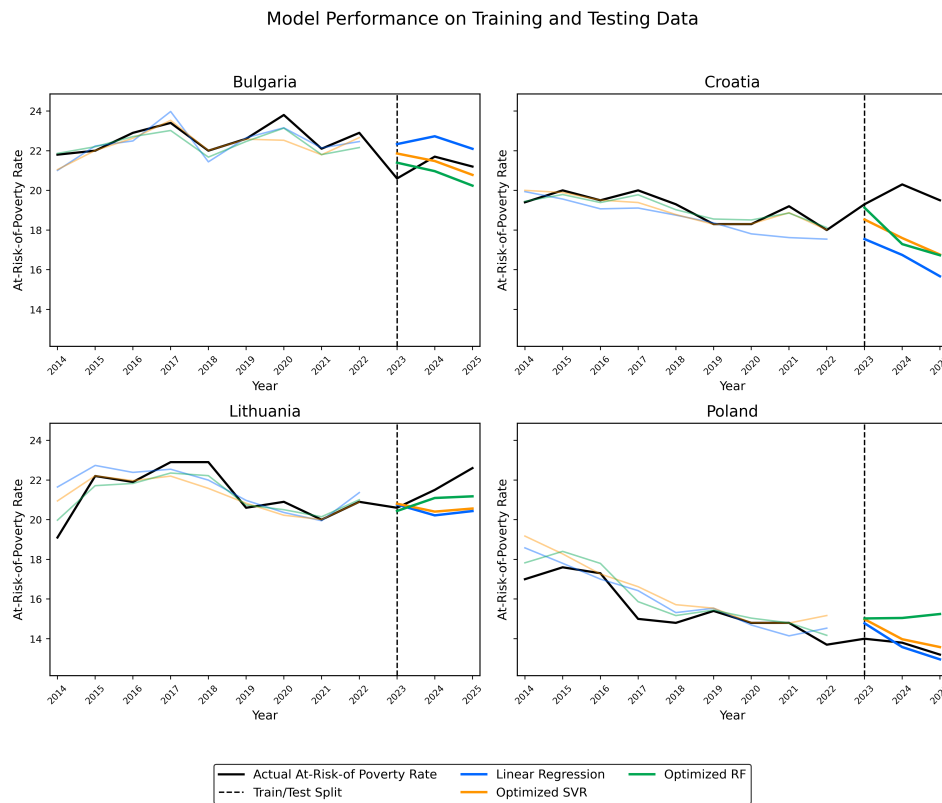


Figure 5: Model performance on training and testing observations by country.

The forecasting analysis demonstrated noticeable differences in predictive behaviour across the analysed countries. Although all models generally captured the direction of the observed poverty trends, forecasting accuracy varied depending on the stability and volatility of the underlying socio-economic conditions within each country. Poland demonstrated the most stable and predictable poverty trajectory during the testing period, while Bulgaria and Croatia exhibited greater year-to-year fluctuations, creating a more challenging forecasting environment for the models.

The train-test comparison further indicated that the implemented models maintained relatively stable predictive behaviour across both periods. Although the training predictions naturally followed the observed poverty trajectories more closely, the models remained comparatively consistent when forecasting future unseen observations. The relatively smooth transition between training and testing forecasts suggests that severe overfitting remained limited despite the comparatively small dataset size.

Across most countries, the optimised Support Vector Regression and optimised Random Forest models followed the observed testing trends more closely than the Linear Regression model. This was particularly visible in countries characterised by greater variation in poverty dynamics, supporting the interpretation that poverty-related socioeconomic relationships contain important non-linear characteristics that are better captured through more flexible machine learning approaches.

The forecasting results for Croatia and Lithuania additionally highlighted the broader socioeconomic context surrounding Eurozone integration. Although the models captured the general direction of poverty developments following euro adoption, the results suggest that poverty-related outcomes remained influenced by multiple interconnected structural factors beyond euro adoption status alone.

Overall, the forecasting and train-test analysis indicate that the implemented machine learning models achieved acceptable generalisation performance and were capable of producing meaningful out-of-sample forecasts within the context of Eastern European socioeconomic data.

4.3 *Explainability and Linear Regression Interpretation*

To improve interpretability and transparency, SHAP (SHapley Additive exPlanations) analysis was applied to the optimised Random Forest model. Figure 6 presents the SHAP summary plot generated from the testing dataset.

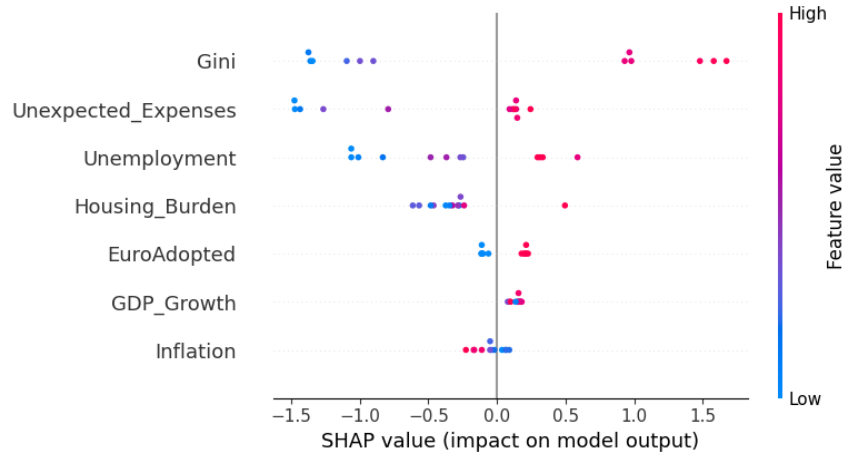


Figure 6: SHAP summary plot for the optimised Random Forest model.

The SHAP analysis identified the Gini coefficient as the most influential predictor of poverty outcomes within the optimised Random Forest model. Higher levels of inequality were consistently associated with increased poverty predictions, supporting the broader interpretation that inequality represents one of the strongest drivers of socio-economic vulnerability within Eastern European countries.

Unexpected expenses and unemployment also demonstrated substantial importance within the model. Higher values for both indicators generally increased predicted poverty outcomes, suggesting that household financial resilience and labour market stability remain closely associated with multidimensional poverty vulnerability.

In contrast, GDP growth and inflation displayed comparatively smaller SHAP contributions overall. Although both variables influenced predictive outcomes, their impact remained weaker than direct indicators of inequality and household financial insecurity. This finding suggests that aggregate macroeconomic indicators alone may be insufficient for explaining poverty-related dynamics without considering broader structural socio-economic conditions.

To complement the SHAP analysis, Figure 7 presents the standardised coefficients obtained from the Linear Regression model.

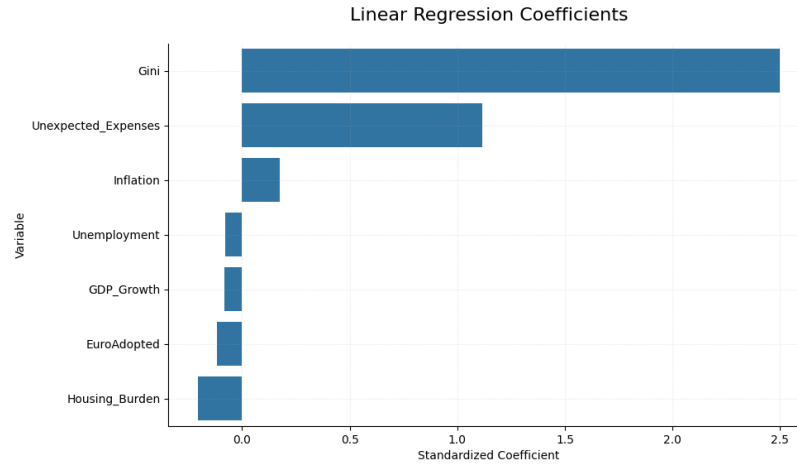


Figure 7: Standardised coefficients from the Linear Regression model.

Although Linear Regression produced weaker predictive performance compared to the non-linear models, the coefficient analysis remained useful for interpretability purposes because it allowed direct examination of the relationship between individual socio-economic variables and poverty predictions.

The coefficient analysis similarly identified the Gini coefficient as the strongest positive contributor to poverty outcomes within the linear framework. Unexpected expenses also demonstrated a strong positive relationship with the at-risk-of-poverty rate, further supporting the interpretation that financial insecurity represents a central component of socio-economic vulnerability.

Inflation and GDP growth displayed comparatively smaller coefficients overall, while housing burden demonstrated a weaker contribution relative to inequality-related indicators. The consistency between the SHAP analysis and the Linear Regression coefficients strengthens the broader interpretation that poverty-related vulnerability within Eastern European economies is influenced more strongly by inequality and household financial resilience than by macroeconomic indicators alone.

Overall, the explainability analysis demonstrates that combining predictive machine learning models with interpretable frameworks such as SHAP can provide valuable insight into the socio-economic variables most strongly associated with poverty-related dynamics.

4.4 Statistical Comparison Before and After Euro Adoption

In addition to the machine learning analysis, independent sample t-tests were conducted in order to examine selected socio-economic indicators

before and after euro adoption in Croatia. Lithuania was excluded from the statistical analysis because only one pre-adoption year was available within the dataset, making meaningful statistical comparison unreliable.

Table 6 presents the independent sample t-test results for Croatia.

Table 6: Independent sample t-test results for Croatia before and after euro adoption

Variable	t-statistic	p-value
At-Risk-of-Poverty Rate	-1.498	0.1950
Inflation	-2.072	0.0921
Unemployment	3.577	0.0059
Gini Coefficient	0.276	0.7980

The statistical analysis identified mixed results regarding socio-economic changes following euro adoption in Croatia. While most analysed indicators did not demonstrate statistically significant differences between the pre-adoption and post-adoption periods, unemployment displayed statistically significant variation at conventional significance levels.

Among the analysed variables, unemployment produced the largest absolute t-statistic together with a statistically significant p-value ($p = 0.0059$). This suggests that labour market conditions changed more substantially between the pre-adoption and post-adoption periods compared to the other analysed indicators. However, the statistical analysis alone does not establish a direct causal relationship between euro adoption and labour market developments, as unemployment trends may also reflect broader economic and structural conditions within Croatia during the observed period.

Inflation additionally demonstrated moderate variation following euro adoption, although the associated p-value remained above conventional significance thresholds. This finding aligns with broader discussions within the literature suggesting that inflation concerns surrounding euro adoption may exist at the household perception level even when statistically significant macroeconomic changes remain more difficult to identify within shorter observational periods.

In contrast, the at-risk-of-poverty rate and Gini coefficient did not demonstrate statistically significant differences following euro adoption. These findings suggest that poverty and inequality dynamics remained influenced by broader structural socio-economic conditions beyond euro adoption status alone.

Overall, the statistical analysis supports the broader interpretation emerging from the machine learning models: poverty-related socio-economic

dynamics within Eastern European economies appear to depend more strongly on multidimensional structural factors such as inequality, labour market stability, and household financial resilience rather than monetary integration alone.

These findings should nevertheless be interpreted cautiously because the available post-adoption period remained relatively short and the dataset size was limited. Consequently, the statistical analysis is intended primarily as a complementary exploratory component supporting the broader machine learning findings rather than as definitive causal evidence regarding the socio-economic effects of euro adoption.

5 DISCUSSION

5.1 *Interpretation of Predictive Performance and Model Comparison*

The findings of this study demonstrate that machine learning approaches can provide meaningful insight into multidimensional poverty-related vulnerability within Eastern European economies undergoing processes of Eurozone integration and economic convergence. The comparatively strong performance of the non-linear models suggests that socioeconomic vulnerability is shaped by interconnected structural relationships that cannot be fully captured through purely linear modelling approaches.

This is socially and economically relevant because poverty-related vulnerability within Eastern European countries is frequently influenced by multiple overlapping forms of economic pressure, including labour market instability, inflation sensitivity, inequality, and limited household financial resilience. Understanding these interactions is particularly important within the context of Eurozone integration, where macroeconomic convergence does not necessarily guarantee equal socioeconomic outcomes at the household level.

Previous literature examining Eurozone integration within Eastern Europe similarly highlighted the persistence of structural economic vulnerability despite broader convergence processes. Heller and Warzala (2019) argued that although Eurozone membership may contribute to macroeconomic stabilization, economic asymmetries and structural vulnerabilities often remain uneven across countries and may become particularly visible during periods of crisis or economic adjustment. Likewise, Panait and Radoi (2025) reported that inflation volatility and financial instability remained significant concerns within non-Euro economies during recent macroeconomic shocks.

The findings of the present study reinforce these broader concerns by demonstrating that variables associated with inequality, inflation sensitivity,

labour market instability, and household financial vulnerability remained strongly connected to at-risk-of-poverty outcomes across both Eurozone and non-Eurozone countries. In particular, the stronger forecasting capability achieved by the optimised Support Vector Regression model suggests that these relationships operate through complex and non-linear structural interactions rather than simple direct associations.

These findings directly answer the second research question of the study concerning how Linear Regression, Support Vector Regression, and Random Forest Regression compare in predictive performance. The results indicate that the non-linear machine learning approaches were more effective for modelling poverty-related socioeconomic dynamics than the traditional linear approach, with the optimised Support Vector Regression model demonstrating the strongest overall predictive capability. This suggests that multidimensional poverty-related vulnerability within Eastern European economies is characterised by structural complexity that is more effectively captured through flexible non-linear modelling techniques.

Compared to the benchmark studies, the present research additionally contributes a methodological extension through the integration of predictive machine learning and explainable artificial intelligence techniques. While previous studies primarily relied on descriptive macroeconomic comparison and convergence analysis, the present study applied machine learning forecasting methods capable of modelling multidimensional socioeconomic relationships across chronological data. The integration of SHAP explainability analysis further strengthened interpretability by identifying the variables most strongly associated with predicted poverty outcomes.

The findings also contribute to broader methodological discussions within socioeconomic machine learning research. Previous poverty prediction studies frequently prioritised predictive accuracy alone, whereas the present study additionally examined forecasting robustness, chronological train-test generalisation, and interpretability within a policy-oriented socioeconomic context. This is particularly important because predictive performance alone may provide limited practical value if the structural drivers behind model predictions remain insufficiently interpretable.

Overall, the findings suggest that combining machine learning forecasting with explainable artificial intelligence techniques may provide a valuable analytical framework for studying multidimensional poverty-related vulnerability within Eastern European economies. More broadly, the results reinforce the argument that socioeconomic exclusion should be understood as a structurally interconnected phenomenon influenced by complex interactions between inequality, inflationary pressure, labour market conditions, and household financial resilience.

5.2 *Socioeconomic Interpretation of Key Indicators*

The explainability analysis demonstrated that inequality, financial vulnerability, and labour market instability were among the most influential predictors of the at-risk-of-poverty rate within the analysed Eastern European countries. In particular, the strong contribution of the Gini coefficient within both the SHAP analysis and the Linear Regression coefficient interpretation suggests that unequal income distribution remains closely associated with broader forms of socioeconomic vulnerability.

These findings are socially significant because inequality influences not only income distribution itself but also broader dimensions of economic resilience, access to opportunities, and exposure to financial insecurity. Households facing greater structural inequality may be more vulnerable to inflationary pressure, unexpected expenses, and labour market instability even during periods of broader macroeconomic growth. Consequently, the findings reinforce multidimensional interpretations of poverty that extend beyond income-based deprivation alone.

The importance of household financial vulnerability was particularly visible through the strong predictive influence of the inability to face unexpected expenses variable. This finding suggests that financial resilience at the household level plays an important role in shaping poverty-related outcomes within Eastern European economies. Even when households remain above formal poverty thresholds, limited financial reserves may still expose individuals to significant socioeconomic insecurity and reduced capacity to absorb economic shocks.

These findings are broadly consistent with existing literature emphasising the structural relationship between inequality, financial insecurity, and socioeconomic exclusion. Previous research argued that poverty-related vulnerability within Eastern European economies remains strongly influenced by labour market conditions, material deprivation, and unequal access to economic resources. Similarly, Heller and Warzala (2019) highlighted the persistence of structural economic asymmetries and uneven socioeconomic adjustment during Eurozone convergence processes, while Panait and Radoi (2025) emphasised the importance of inflation sensitivity and financial instability within countries exposed to broader macroeconomic shocks.

The present findings extend these benchmark studies by demonstrating through machine learning explainability analysis that variables associated with inequality and financial vulnerability remained among the strongest predictors within the implemented forecasting models. In particular, the explainability analysis suggests that direct indicators of household-level

economic stress may provide stronger insight into poverty-related vulnerability than broader aggregate macroeconomic indicators alone.

Unemployment additionally demonstrated substantial influence within the predictive models, reinforcing broader literature identifying labour market instability as a major contributor to socioeconomic insecurity. The statistical comparison conducted for Croatia also identified unemployment as the only variable demonstrating statistically significant variation between the pre-adoption and post-adoption periods. Although the results should be interpreted cautiously due to the relatively short observational timeframe, the findings nevertheless support existing concerns regarding labour market vulnerability during periods of economic transition and integration discussed within the benchmark literature.

The comparatively weaker contribution of GDP growth within the explainability analysis is also noteworthy. Although economic growth remains important for broader macroeconomic development, the findings suggest that aggregate growth alone may not sufficiently explain variations in household-level socioeconomic vulnerability. This observation aligns with the conclusions of Heller and Warzala (2019), who argued that macroeconomic convergence may coexist with persistent structural inequalities and uneven socioeconomic resilience across countries undergoing Eurozone integration.

These findings directly answer the first research question of the study, which examined which socioeconomic indicators are most strongly associated with the at-risk-of-poverty rate in Eastern European countries. The results indicate that inequality, financial vulnerability, unemployment, and inflation-related indicators represented some of the most influential predictors of poverty-related outcomes within the implemented machine learning models.

At the same time, the findings also address the third research question concerning how inequality, inflation, and financial vulnerability are associated with poverty-related outcomes in the context of Eurozone integration. The results suggest that poverty-related vulnerability within Eastern European economies remained strongly connected to broader structural conditions such as financial resilience, labour market stability, and inequality rather than Eurozone membership status alone.

Overall, the findings reinforce the argument that poverty should be understood as a multidimensional and structurally interconnected phenomenon. The results suggest that socioeconomic vulnerability within Eastern European economies is influenced less by isolated macroeconomic indicators alone and more by the interaction between inequality, labour market instability, inflation sensitivity, and household financial resilience.

5.3 *Explainability and Model Transparency*

One of the central contributions of this study concerns the integration of explainable artificial intelligence techniques within socioeconomic prediction research. Although advanced machine learning models frequently achieve strong predictive performance, existing literature has highlighted concerns regarding limited interpretability and “black-box” behaviour within complex predictive systems (Lundberg & Lee, 2017). Within policy-oriented socioeconomic research, this represents an important challenge because predictive outcomes alone may provide limited practical value if the structural mechanisms influencing those predictions remain insufficiently transparent.

The integration of SHAP explainability analysis within the present study substantially improved model transparency by identifying the variables most strongly associated with predicted poverty outcomes and illustrating both the direction and relative magnitude of their influence. This made it possible to interpret how inequality, financial vulnerability, unemployment, and inflation-related indicators contributed to the behaviour of the implemented machine learning models.

These findings are particularly important within the context of socioeconomic policymaking because poverty-related vulnerability is closely connected to inequality, labour market insecurity, and household financial resilience. Existing literature similarly emphasised that explainability is especially relevant in policy-oriented research contexts where understanding model behaviour remains essential for interpretation and decision-making (Lundberg & Lee, 2017).

The findings additionally reinforce broader concerns within machine learning literature regarding the interpretability limitations of advanced predictive systems (Lundberg & Lee, 2017). Previous socioeconomic prediction studies frequently prioritised predictive performance while providing comparatively limited explanation regarding model behaviour and variable interaction. In contrast, the present study integrated explainability directly into the forecasting framework through the application of SHAP analysis alongside Linear Regression coefficient interpretation.

Compared to benchmark studies examining Eurozone integration and macroeconomic convergence, the present study contributes not only predictive modelling capability but also an additional interpretive dimension. While Heller and Warzala (2019) and Panait and Radoi (2025) primarily relied on descriptive macroeconomic comparison and structural interpretation, the present research demonstrates how explainable artificial intelligence techniques can provide more detailed insight into the

multidimensional socioeconomic relationships underlying poverty-related vulnerability.

The consistency observed between the SHAP analysis and the Linear Regression coefficient interpretation additionally strengthened the credibility of the findings. Although the Linear Regression model achieved comparatively weaker predictive performance, both approaches identified inequality, labour market instability, and financial vulnerability as some of the most influential structural drivers of poverty-related outcomes within the analysed countries.

These findings directly answer the fourth research question concerning how SHAP explainability methods can improve the interpretability of machine learning models in socioeconomic research. The results demonstrate that SHAP substantially improved transparency by allowing detailed interpretation of model behaviour while simultaneously preserving the strong predictive capability of the non-linear machine learning models.

More broadly, the findings suggest that explainable artificial intelligence techniques may play an increasingly important role within applied socioeconomic forecasting research. Combining predictive machine learning models with transparent interpretability frameworks may improve both methodological robustness and policy relevance, particularly within research contexts involving multidimensional socioeconomic vulnerability and economic integration processes (Lundberg & Lee, 2017).

Overall, the findings demonstrate that strong predictive performance and interpretability do not necessarily need to exist in opposition. Instead, the integration of explainable artificial intelligence techniques can help bridge the gap between advanced predictive modelling and transparent socioeconomic interpretation.

5.4 *Forecasting Robustness and Generalisation Performance*

An important objective of the present study involved evaluating whether the implemented machine learning models could maintain reliable predictive performance when applied to unseen chronological data. This is particularly relevant within socioeconomic forecasting research because models trained on historical data may perform well during training while failing to generalise effectively across later periods characterised by different economic conditions or structural changes.

The chronological train-test split implemented within the present study therefore provided a more realistic evaluation of forecasting robustness by assessing model performance on future observations rather than randomly shuffled data.

The forecasting results demonstrated that all implemented models maintained relatively stable predictive capability across the testing period, although differences in generalisation performance remained observable. The optimised Support Vector Regression model achieved the strongest balance between predictive accuracy and forecasting stability, suggesting that non-linear machine learning approaches may be particularly effective for modelling socioeconomic relationships characterised by temporal variation and structural complexity.

The forecasting visualisations additionally demonstrated that the implemented models generally preserved broader poverty-related trajectories across the analysed countries despite moderate divergence during periods of greater volatility. These findings are important because socioeconomic vulnerability within Eastern European economies is influenced by evolving structural conditions, inflationary pressure, labour market fluctuations, and economic adjustment processes that may vary across countries and time periods.

The comparatively stable forecasting behaviour observed in the optimised Support Vector Regression and Random Forest models reinforces previous literature arguing that advanced machine learning approaches are capable of capturing complex interactions within multidimensional socioeconomic datasets ([breiman_random_2001](#); Usmanova et al., [2022](#); Zixi, [2021](#)). At the same time, the forecasting analysis demonstrated that model robustness remains closely connected to appropriate validation procedures and model optimisation strategies.

The findings additionally highlight the methodological importance of chronological validation within applied socioeconomic forecasting research. Random train-test splitting approaches may artificially inflate predictive performance when temporal dependencies exist within the data. By contrast, the chronological evaluation framework implemented within the present study provided a more conservative assessment of forecasting capability under evolving socioeconomic conditions.

Compared to benchmark studies examining Eurozone integration and macroeconomic convergence, the present study contributes an additional forecasting dimension through chronological machine learning validation and predictive robustness analysis. While Heller and Warzala (2019) and Panait and Radoi (2025) primarily focused on descriptive macroeconomic comparison, the present research examined whether machine learning models could generalise across later periods characterised by changing socioeconomic conditions.

These findings directly answer the fifth research question concerning how robust the implemented machine learning models remained when evaluated on unseen time-series data. The results suggest that the non-

linear machine learning approaches demonstrated comparatively strong generalisation capability despite the relatively limited dataset size, particularly following Bayesian hyperparameter optimisation.

Nevertheless, some forecasting deviations remained observable across selected country-level trajectories, particularly during periods characterised by greater volatility and structural fluctuation. This suggests that although machine learning models can provide meaningful forecasting capability within socioeconomic research, predictive performance may still be influenced by dataset size, temporal coverage, and broader macroeconomic instability.

Overall, the findings indicate that machine learning forecasting approaches can provide relatively robust predictive performance within multidimensional socioeconomic research contexts when appropriate chronological validation and model optimisation procedures are implemented.

5.5 *Eurozone Integration and Broader Socioeconomic Implications*

The findings of this study suggest that poverty-related socioeconomic vulnerability within Eastern European economies cannot be explained through Eurozone integration or macroeconomic convergence indicators alone. Instead, broader structural conditions such as inequality, labour market instability, inflation sensitivity, and household financial vulnerability remained more strongly associated with at-risk-of-poverty outcomes across the analysed countries.

These findings are particularly relevant within the context of ongoing economic integration because public debates surrounding euro adoption frequently focus on inflation, purchasing power, affordability, and household-level economic security. While convergence criteria primarily evaluate inflation stability, fiscal discipline, and monetary integration, the present findings suggest that these measures alone provide only limited insight into multidimensional socioeconomic vulnerability at the household level.

The statistical comparison conducted for Croatia further reinforces this interpretation. Although moderate variation remained observable across several socioeconomic indicators following euro adoption, unemployment was the only variable demonstrating statistically significant change during the analysed period. Poverty-related vulnerability nevertheless remained strongly associated with broader structural indicators linked to inequality and financial resilience, suggesting that the socioeconomic effects of Eurozone integration may remain uneven and gradual.

The findings are broadly consistent with previous literature examining Eurozone convergence and structural economic vulnerability within

Eastern Europe. Heller and Warzala (2019) argued that although Eurozone membership may contribute to macroeconomic stabilization, structural asymmetries and uneven economic adjustment frequently persist across countries undergoing convergence processes. Similarly, Panait and Radoi (2025) highlighted the importance of inflation volatility and financial instability within economies exposed to broader macroeconomic shocks.

The present findings reinforce these benchmark studies by demonstrating that poverty-related vulnerability remained closely connected to household financial resilience, labour market conditions, and inequality even within countries participating in broader convergence processes. In particular, the comparatively weaker contribution of GDP growth within the explainability analysis suggests that aggregate macroeconomic improvement alone may not necessarily translate into equivalent reductions in household-level socioeconomic vulnerability.

These findings additionally address the third research question concerning how inequality, inflation, and financial vulnerability are associated with poverty-related outcomes in the context of Eurozone integration. The results indicate that these structural socioeconomic indicators remained strongly connected to predicted poverty outcomes across both Eurozone and non-Eurozone countries.

More broadly, the findings also contribute to answering the main research question concerning the extent to which machine learning models can predict the at-risk-of-poverty rate using socioeconomic indicators within Eastern European economies in the context of Eurozone integration. The results demonstrate that machine learning approaches were capable of effectively modelling poverty-related vulnerability through multidimensional socioeconomic indicators while additionally identifying the structural variables most strongly associated with predicted outcomes.

Compared to existing literature focused primarily on descriptive macroeconomic comparison and convergence analysis, the present study demonstrates the additional value of integrating predictive machine learning and explainable artificial intelligence techniques within the analysis of socioeconomic vulnerability. The findings suggest that combining forecasting capability with interpretability may provide more detailed insight into the structural drivers of poverty-related outcomes during processes of economic integration and adjustment.

Overall, the findings reinforce the argument that poverty-related vulnerability within Eastern European economies should be understood as a multidimensional and structurally interconnected phenomenon. While Eurozone integration may contribute to broader macroeconomic stabilization, the results suggest that inequality, labour market instability, inflation

sensitivity, and household financial resilience remain central determinants of socioeconomic vulnerability within the analysed countries.

5.6 *Limitations and Future Research Directions*

Several limitations should be acknowledged when interpreting the findings of the present study. First, the analysis relied on a comparatively limited dataset covering four Eastern European countries between 2014 and 2025. Although the implemented machine learning models achieved relatively strong predictive performance, larger longitudinal datasets covering additional countries and longer time periods could improve robustness and allow broader analysis of socioeconomic change.

The selected timeframe was constrained by data availability, as 2014–2025 represented the only period for which sufficiently consistent socioeconomic data covering all analysed countries and indicators remained available simultaneously. Earlier periods would have introduced substantial missing data and reduced comparability between observations.

The timeframe additionally created uneven pre-adoption and post-adoption coverage across countries. Lithuania had only one pre-adoption year available, while Croatia had only two post-adoption years following euro adoption. Consequently, the statistical comparison component of the study was conducted only for Croatia and should be interpreted as exploratory rather than definitive.

The findings should also be interpreted within the broader methodological limitations associated with poverty measurement itself. Existing literature emphasises that formal poverty indicators may not fully capture wider forms of material deprivation and household financial insecurity (Van De Werfhorst & Salverda, 2012; Whelan & Maître, 2012). Households may remain above official poverty thresholds while still experiencing financial vulnerability, housing burden, or inability to absorb unexpected expenses. The predictive importance of variables related to unexpected expenses, inequality, and labour market instability further supports multi-dimensional poverty literature (Usmanova et al., 2022; Whelan & Maître, 2012; Zixi, 2021).

Future research could extend the present study through larger longitudinal datasets covering additional countries and socioeconomic contexts. Applying the implemented machine learning and explainability framework to other economies could provide further insight into whether the identified relationships remain consistent across different economic systems.

Another important direction for future research concerns Bulgaria following its Eurozone accession in 2026. Within the present study, Bulgaria primarily functioned as a non-Eurozone comparison country. Future stud-

ies could therefore investigate whether the machine learning framework developed in this research would accurately predict poverty-related socioeconomic outcomes during Bulgaria's post-adoption period.

Finally, future research could further examine the relationship between household financial resilience, inflation sensitivity, and socioeconomic vulnerability during periods of economic integration and macroeconomic adjustment, particularly within Eastern European economies.

6 CONCLUSION

This study examined the extent to which machine learning models can predict the at-risk-of-poverty rate using socioeconomic indicators within Eastern European countries in the context of Eurozone integration. By combining machine learning forecasting methods with explainable artificial intelligence techniques, the research analysed multidimensional socioeconomic relationships associated with inequality, unemployment, inflation, financial vulnerability, and material deprivation across Bulgaria, Croatia, Lithuania, and Poland.

The findings demonstrated that non-linear machine learning approaches, particularly the optimised Support Vector Regression model, were capable of effectively modelling poverty-related socioeconomic dynamics within the analysed countries. The results additionally highlighted the importance of inequality, labour market instability, inflation sensitivity, and household financial resilience as some of the strongest predictors associated with poverty-related vulnerability.

Another important contribution of the study concerns the integration of explainable artificial intelligence within socioeconomic forecasting research. Through the application of SHAP explainability analysis, the study demonstrated that machine learning models can provide not only strong predictive capability but also interpretable insight into the structural socioeconomic variables influencing poverty-related outcomes. This improved the transparency and policy relevance of the implemented analytical framework.

The findings further suggest that macroeconomic convergence and Eurozone integration alone may not fully eliminate household-level socioeconomic vulnerability if broader structural inequalities and financial insecurity persist. Consequently, the study reinforces the importance of multidimensional approaches to poverty analysis that extend beyond aggregate macroeconomic indicators alone.

Overall, the research contributes to growing interdisciplinary literature combining data science, economics, and public policy by demonstrating how machine learning and explainable artificial intelligence techniques

can support the analysis of multidimensional socioeconomic vulnerability within Eastern European economies. The study additionally highlights the importance of maintaining transparency and interpretability when applying advanced predictive systems within socially sensitive policy-oriented research contexts.

ACKNOWLEDGEMENTS

I would like to thank the members of the thesis committee for their time and evaluation of this work. I would like to express my sincere gratitude to my thesis supervisor, Dr. Ir. Federico Zamberlan, for his guidance, feedback, and support throughout the development of this thesis. His insights and supervision were invaluable throughout the research and writing process.

Finally, I would like to thank my family and friends who supported me throughout my studies and during the completion of this thesis.

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